

OUTPUT:

1 . Opening Git Bash and creating a Python file (sjf.py) using nano editor to implement SJF preemptive scheduling program.



```
MINGW64 ~/C:/Users/VICTUS  
VICTUS@AAM MINGW64 ~ (master)  
$ nano sjf.py
```

2. Writing the Python program for Shortest Job First (Preemptive) scheduling algorithm to calculate total context switches and average waiting time.

```
MINGW64/C:/Users/VICTOR
GNU nano 8.7
bt=[10,20,30]
at=[0,2,6]

rt=bt[:]
t=0
complete=0
prev=-1
cs=-1
wt=[0]*3

while complete!=3:
    shortest=-1
    minm=9999
    for i in range(3):
        if at[i]<=t and rt[i]>0 and rt[i]<minm:
            minm=rt[i]
            shortest=i

    if shortest==-1:
        t+=1
        continue

    if prev!=shortest:
        cs+=1
        prev=shortest

    rt[shortest]-=1
    t+=1

    if rt[shortest]==0:
        complete+=1
        wt[shortest]=t-bt[shortest]-at[shortest]

print("Total Context Switches =",cs)
print("Average Waiting Time =",round(sum(wt)/3,2),"ns")
```

3. Saving the program and executing it in Git Bash using the command `python sjf.py`.

```
MINGW64:/c/Users/VICTUS  
VICTUS@AAM MINGW64 ~ (master)  
$ nano sjf.py  
VICTUS@AAM MINGW64 ~ (master)  
$ python sjf.py
```

4. *Output of the program showing total context switches and average waiting time for the given processes.*

 MINGW64:/c/Users/VICTUS

VICTUS@AAM MINGW64 ~ (master)

\$ nano sjf.py

VICTUS@AAM MINGW64 ~ (master)

\$ python sjf.py

Total Context Switches = 2

Average Waiting Time = 10.67 ns

VICTUS@AAM MINGW64 ~ (master)

\$ |